

Duality in Category Theory

Leopold Wuhan Zhou
School of EECS, Peking University

July 23, 2026

Abstract

This note is based on lectures by Eugenia Cheng at the University of Chicago in summer 2026. We introduce opposite categories and explain how categorical duality exchanges initial and terminal objects, products and coproducts, and monomorphisms and epimorphisms. The universal properties and the basic examples are stated and proved carefully.

Contents

1	Examples	2
2	Categories and Opposite Categories	2
3	Initial and Terminal Objects	4
4	Products and Coproducts	4
5	Monomorphisms and Epimorphisms	5

1 Examples

Many mathematical notions naturally occur in dual pairs. Familiar examples include

lcm	$\longleftrightarrow$	gcd
max	$\longleftrightarrow$	min
$\cap$	$\longleftrightarrow$	$\cup$
sup	$\longleftrightarrow$	inf
injective	$\longleftrightarrow$	surjective
Cartesian product	$\longleftrightarrow$	disjoint union
$\times$	$\longleftrightarrow$	$+$
$\geq$	$\longleftrightarrow$	$\leq$
$\emptyset$	$\longleftrightarrow$	$\{*\}$.

Category theory makes this phenomenon precise: reversing every arrow in a definition or theorem produces its *dual*.

2 Categories and Opposite Categories

Definition 2.1. A *category* $\mathcal{C}$ consists of the following data.

1. A class $\text{Ob}(\mathcal{C})$ of objects.
2. For every $A, B \in \text{Ob}(\mathcal{C})$, a set $\text{Hom}_{\mathcal{C}}(A, B)$ of morphisms from A to B .
3. For every object A , an identity morphism $\text{id}_A: A \rightarrow A$.
4. For morphisms $f: A \rightarrow B$ and $g: B \rightarrow C$, a composite $g \circ f: A \rightarrow C$.

These data satisfy:

$$\begin{aligned} h \circ (g \circ f) &= (h \circ g) \circ f && \text{whenever both sides are defined,} \\ \text{id}_B \circ f &= f = f \circ \text{id}_A && \text{for every } f: A \rightarrow B. \end{aligned}$$

Remark 2.2. Allowing a class of objects accommodates large categories such as **Set**. The requirement that each collection of morphisms be a set says that $\mathcal{C}$ is *locally small*.

Definition 2.3. Given a category $\mathcal{C}$, its *opposite category* $\mathcal{C}^{\text{op}}$ is defined by

$$\text{Ob}(\mathcal{C}^{\text{op}}) = \text{Ob}(\mathcal{C}), \quad \text{Hom}_{\mathcal{C}^{\text{op}}}(A, B) = \text{Hom}_{\mathcal{C}}(B, A).$$

The identity morphisms are unchanged. If $f^{\text{op}}: A \rightarrow B$ and $g^{\text{op}}: B \rightarrow C$ correspond respectively to $f: B \rightarrow A$ and $g: C \rightarrow B$ in $\mathcal{C}$, then

$$g^{\text{op}} \circ_{\mathcal{C}^{\text{op}}} f^{\text{op}} := (f \circ_{\mathcal{C}} g)^{\text{op}}.$$

Proposition 2.4. $\mathcal{C}^{\text{op}}$ is a category, and $(\mathcal{C}^{\text{op}})^{\text{op}} = \mathcal{C}$.

Proof. The unit laws in $\mathcal{C}^{\text{op}}$ follow from those in $\mathcal{C}$. Associativity follows because reversing a string of arrows reverses its order, while composition in $\mathcal{C}$ is associative. Reversing all arrows twice restores the original hom-sets and composition. $\square$

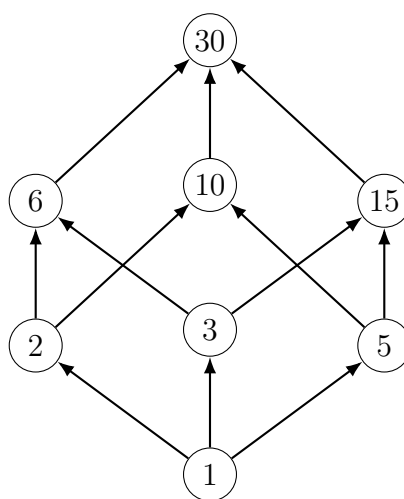
Remark 2.5. Every theorem expressed purely in the language of objects, morphisms, identities, and composition has a dual theorem obtained by reversing all morphisms and reversing the order of composition.

Example 2.6. Every partially ordered set $(P, \leq)$ determines a category: its objects are the elements of P , and there is a unique morphism $x \rightarrow y$ exactly when $x \leq y$. Reflexivity gives identities and transitivity gives composition. Such a category is called *thin*, since every hom-set has at most one element. Its opposite category is the poset $(P, \geq)$. For example,

$$0 \longrightarrow 1 \longrightarrow 2 \longrightarrow 3 \longrightarrow \cdots \quad \text{becomes} \quad 0 \longleftarrow 1 \longleftarrow 2 \longleftarrow 3 \longleftarrow \cdots .$$

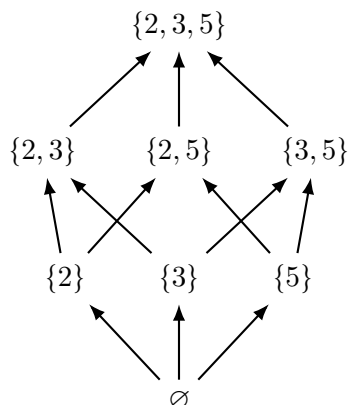
Example 2.7. The objects of **Set** are sets and its morphisms are functions. Thus a morphism $A \rightarrow B$ in **Set**^{op} is, by definition, an ordinary function $B \rightarrow A$. Notice that this does not usually make **Set**^{op} equivalent to **Set**; the reversal is formal.

Example 2.8. Order the positive divisors of 30 by divisibility. A morphism $a \rightarrow b$ means $a \mid b$. The Hasse diagram (arrows show covering relations) is



The opposite category reverses every arrow, so a morphism $a \rightarrow b$ in the opposite category asserts that $b \mid a$.

Example 2.9. For a set S , the power set $\mathcal{P}(S)$ is a category under inclusion: there is a unique morphism $A \rightarrow B$ precisely when $A \subseteq B$. Its opposite has a morphism $A \rightarrow B$ precisely when $A \supseteq B$. For $S = \{2, 3, 5\}$, the inclusion category is the Boolean lattice



and the opposite category has the same diagram with all arrows reversed.

3 Initial and Terminal Objects

Definition 3.1. An object I of a category $\mathcal{C}$ is *initial* if for every object X there exists a unique morphism $I \rightarrow X$. An object T is *terminal* if for every object X there exists a unique morphism $X \rightarrow T$.

Remark 3.2. An object is terminal in $\mathcal{C}$ if and only if it is initial in $\mathcal{C}^{\text{op}}$. Thus initial and terminal objects are dual notions.

Example 3.3. In **Set**, the empty set is initial: for every set X there is exactly one function $\emptyset \rightarrow X$. Every singleton $\{*\}$ is terminal: there is exactly one function $X \rightarrow \{*\}$.

Theorem 3.4 (Uniqueness up to unique isomorphism). *If I and I' are initial objects of $\mathcal{C}$, then there is a unique isomorphism $I \cong I'$. Dually, any two terminal objects are uniquely isomorphic.*

Proof. Initiality gives unique morphisms $f: I \rightarrow I'$ and $g: I' \rightarrow I$. Both $g \circ f$ and id_I are morphisms $I \rightarrow I$; by initiality of I , they are equal. Similarly, $f \circ g = \text{id}_{I'}$. Hence f and g are inverse isomorphisms. Since there is only one morphism $I \rightarrow I'$, the isomorphism is unique. The terminal statement is the dual argument. $\square$

Example 3.5. In **Grp**, the trivial group 1 is both initial and terminal. Indeed, there is a unique group homomorphism $1 \rightarrow G$ and a unique group homomorphism $G \rightarrow 1$ for every group G . An object that is both initial and terminal is called a *zero object*.

4 Products and Coproducts

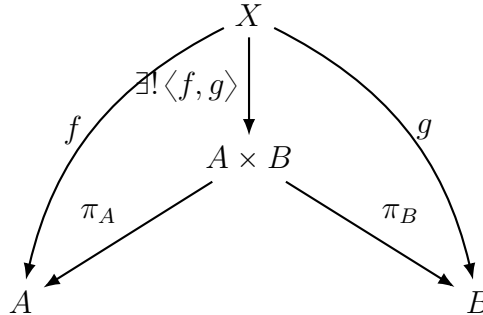
Definition 4.1. Let A, B be objects of a category $\mathcal{C}$. A *product* of A and B is an object $A \times B$ together with projection morphisms

$$A \xleftarrow{\pi_A} A \times B \xrightarrow{\pi_B} B$$

such that for every object X and every pair of morphisms $f: X \rightarrow A$ and $g: X \rightarrow B$, there exists a unique morphism $\langle f, g \rangle: X \rightarrow A \times B$ for which

$$\pi_A \circ \langle f, g \rangle = f, \quad \pi_B \circ \langle f, g \rangle = g.$$

Equivalently, the following diagram commutes:



Remark 4.2. Fixing A and B , form the category of cones over the discrete diagram consisting of A and B . Its objects are triples (X, f, g) with $f: X \rightarrow A$ and $g: X \rightarrow B$; a morphism $(X, f, g) \rightarrow (Y, f', g')$ is a morphism $h: X \rightarrow Y$ such that $f' \circ h = f$ and $g' \circ h = g$. A product is precisely a terminal object in this cone category. This explains both the universal property and the uniqueness theorem below.

Proposition 4.3. *If a product of A and B exists, then it is unique up to a unique isomorphism commuting with the two projections.*

Proof. Suppose (P, p_A, p_B) and (Q, q_A, q_B) are products. Universality gives unique morphisms $u: P \rightarrow Q$ and $v: Q \rightarrow P$ preserving both projections. The morphisms $v \circ u$ and id_P have the same composites with p_A and p_B , so uniqueness in the universal property of P gives $v \circ u = \text{id}_P$. Similarly $u \circ v = \text{id}_Q$. The morphism u is therefore the unique projection-preserving isomorphism. $\square$

Definition 4.4. A *coproduct* of A and B in $\mathcal{C}$ is a product of A and B in $\mathcal{C}^{\text{op}}$. Explicitly, it is an object $A \sqcup B$ with injections

$$A \xrightarrow{\iota_A} A \sqcup B \xleftarrow{\iota_B} B$$

such that for every $f: A \rightarrow X$ and $g: B \rightarrow X$, there is a unique morphism $[f, g]: A \sqcup B \rightarrow X$ satisfying

$$[f, g] \circ \iota_A = f, \quad [f, g] \circ \iota_B = g.$$

Example 4.5. In **Set**, the categorical product is the Cartesian product

$$A \times B = \{(a, b) : a \in A, b \in B\},$$

with coordinate projections. The categorical coproduct is the disjoint union, which can be realized without assuming A and B disjoint as

$$A \sqcup B = (A \times \{0\}) \cup (B \times \{1\}).$$

Example 4.6. In a poset regarded as a category, a product is a greatest lower bound (meet) and a coproduct is a least upper bound (join), when they exist. Consequently, in the divisibility category $(\mathbb{N}_{>0}, |)$,

$$a \times b = \gcd(a, b), \quad a \sqcup b = \text{lcm}(a, b).$$

In the inclusion category $\mathcal{P}(S)$, products are intersections and coproducts are unions.

5 Monomorphisms and Epimorphisms

Elements of a set X can be encoded categorically: each $x \in X$ corresponds to the function $\bar{x}: \{*\} \rightarrow X$ with $\bar{x}(*) = x$. This observation motivates the following definitions, which make sense in every category.

Definition 5.1. A morphism $f: A \rightarrow B$ in $\mathcal{C}$ is a *monomorphism* (or *monic*) if for every object X and all morphisms $s, t: X \rightarrow A$,

$$f \circ s = f \circ t \implies s = t.$$

A morphism $f: A \rightarrow B$ is an *epimorphism* (or *epic*) if for every object Y and all morphisms $s, t: B \rightarrow Y$,

$$s \circ f = t \circ f \implies s = t.$$

Remark 5.2. A morphism is epic in $\mathcal{C}$ if and only if the corresponding reversed morphism is monic in $\mathcal{C}^{\text{op}}$. Thus monomorphisms and epimorphisms are dual notions.

Proposition 5.3. *In $\mathbf{Set}$, the monomorphisms are exactly the injective functions.*

Proof. Let $f: A \rightarrow B$ be injective. If $f \circ s = f \circ t$ for functions $s, t: X \rightarrow A$, then for each $x \in X$, $f(s(x)) = f(t(x))$; injectivity gives $s(x) = t(x)$. Hence $s = t$.

Conversely, suppose f is not injective. Choose distinct $a_1, a_2 \in A$ with $f(a_1) = f(a_2)$. Define $s, t: \{*\} \rightarrow A$ by $s(*) = a_1$ and $t(*) = a_2$. Then $s \neq t$ but $f \circ s = f \circ t$, so f is not monic. $\square$

Proposition 5.4. *In $\mathbf{Set}$, the epimorphisms are exactly the surjective functions.*

Proof. Let $f: A \rightarrow B$ be surjective. If $s, t: B \rightarrow Y$ satisfy $s \circ f = t \circ f$, then for each $b \in B$ choose $a \in A$ with $f(a) = b$. It follows that $s(b) = s(f(a)) = t(f(a)) = t(b)$, so $s = t$.

Conversely, suppose f is not surjective, and choose $b_0 \in B \setminus f(A)$. Define $s, t: B \rightarrow \{0, 1\}$ by

$$s(b) = 0 \quad \text{for every } b \in B, \quad t(b) = \begin{cases} 1, & b = b_0, \\ 0, & b \neq b_0. \end{cases}$$

Then $s \neq t$, but $s \circ f = t \circ f$ because b_0 is not in the image of f . Therefore f is not epic. $\square$